ENTERPRISE NETWORK INFRASTRUCTURE PROJECT

Dual-ISP Redundancy | OSPF | HSRP | VLANs | NAT (PAT) | DHCP | SSH

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1. Executive Summary

This project presents the design and implementation of a secure, scalable, and highly available enterprise network built in Cisco Packet Tracer. The network follows Cisco's three-layer hierarchical model, consisting of Core, Distribution, and Access layers.

To eliminate single points of failure, the infrastructure implements dual-ISP redundancy, OSPF dynamic routing, HSRP gateway redundancy, PAT (NAT overload), centralized DHCP services, VLAN segmentation, and SSH version 2 for secure remote administration.

The project demonstrates enterprise-level networking concepts that are commonly deployed in modern organizations, and is suitable as a CCNA capstone, portfolio, or coursework submission.

2. Introduction

Enterprise organizations require reliable and scalable networks that support continuous business operations, since downtime can lead to financial loss and reduced productivity.

This project simulates a real enterprise campus network using Cisco technologies. The design focuses on redundancy, security, scalability, and centralized network management, and was built and validated entirely within Cisco Packet Tracer.

3. Project Objectives

- Design a hierarchical enterprise campus network (Core, Distribution, Access).
- Implement VLAN segmentation across all departments.
- Configure Inter-VLAN routing on the distribution-layer multilayer switches.
- Deploy OSPF dynamic routing between all Layer 3 devices.
- Configure HSRP for default-gateway redundancy.
- Provide dual-ISP Internet connectivity for WAN redundancy.
- Implement PAT (NAT overload) for Internet access.
- Configure centralized DHCP services with DHCP relay.
- Enable secure SSH version 2 remote management.
- Improve overall network availability and fault tolerance.

4. Network Requirements

The organization consists of six departments:

- Sales and Marketing
- Human Resources and Logistics
- Finance and Accounts
- Administration and Public Relations
- ICT Department
- Server Room

The network must provide the following capabilities:

- Secure communication between departments

- High availability at the WAN and distribution layers
- Dynamic routing with fast convergence
- Redundant Internet access
- Automatic failover for gateways and WAN links
- Centralized IP address management

5. Network Architecture

The network follows Cisco's three-layer hierarchical model:

Core Layer

- Two Cisco 2811 routers (Router3, Router4)
- Dual-ISP WAN connectivity with cross-connected uplinks for redundancy

Distribution Layer

- Two Cisco 3560 multilayer switches (Multilayer Switch0, Multilayer Switch2)

Responsibilities of the distribution layer:

- Inter-VLAN routing
- HSRP default-gateway redundancy
- OSPF adjacency toward the core routers

Access Layer

Six Cisco 2960 access switches — one per department — connect end devices including PCs, printers, wireless access points, laptops, smartphones, and servers.

6. Network Topology

Figure 1 below shows the complete logical topology, including the dual-ISP WAN edge, the cross-connected core/distribution links, the six department VLANs, and the server-room subnet.

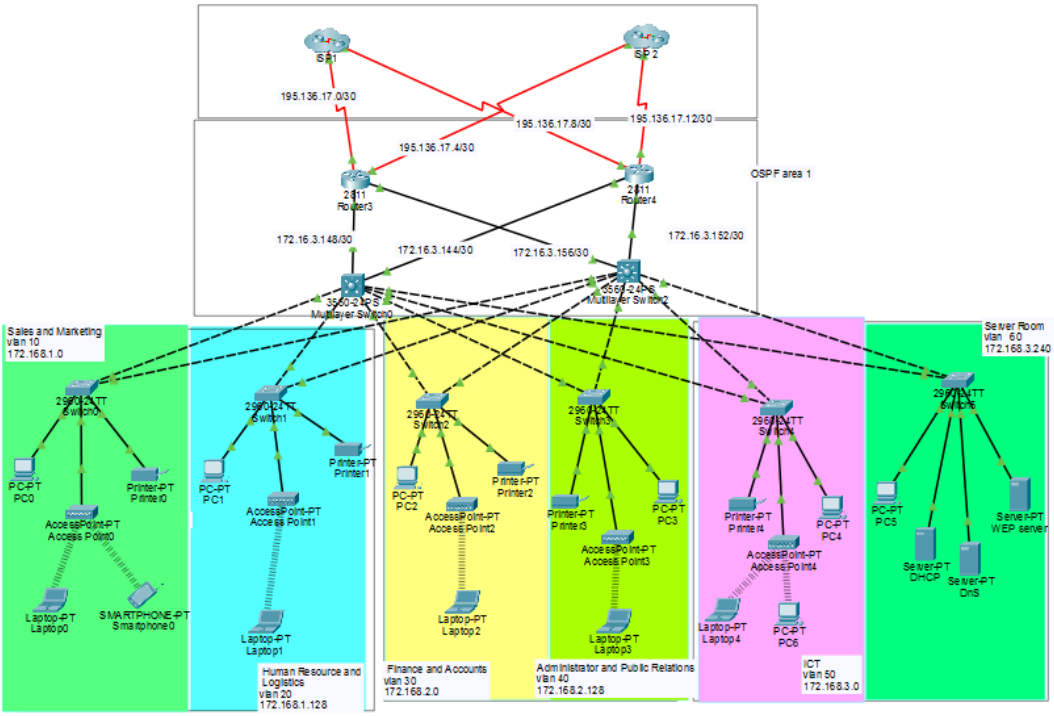

Figure 1. Enterprise Network Topology

7. IP Addressing Plan

The design uses the RFC 1918 private range 172.16.0.0/16, subnetted with a variable-length subnet mask (VLSM) to fit each department's host requirements. Note: the topology diagram labels the department subnets as “172.168.x.x”; this is corrected below to “172.16.x.x”, since 172.168.0.0/16 does not fall inside a valid RFC 1918 private block (the private 172.x range only spans 172.16.0.0 – 172.31.255.255), while 172.16.x.x is a valid, routable-within-the-enterprise private network.

7.1 LAN / VLAN Subnets

VLAN	Department	Network	Usable Range	Broadcast
10	Sales & Marketing	172.16.1.0/25	172.16.1.1 – 172.16.1.126	172.16.1.127
20	HR & Logistics	172.16.1.128/25	172.16.1.129 – 172.16.1.254	172.16.1.255
30	Finance & Accounts	172.16.2.0/25	172.16.2.1 – 172.16.2.126	172.16.2.127
40	Administration & PR	172.16.2.128/25	172.16.2.129 – 172.16.2.254	172.16.2.255
50	ICT	172.16.3.0/25	172.16.3.1 – 172.16.3.126	172.16.3.127
60	Server Room	172.16.3.240/28	172.16.3.241 – 172.16.3.254	172.16.3.255

7.2 Core-to-Distribution Links (OSPF Area 1)

Each core router is dual-homed to both distribution switches, giving four /30 point-to-point transit links:

Link	Network
Router3 ↔ Multilayer Switch0	172.16.3.148/30
Router3 ↔ Multilayer Switch2	172.16.3.144/30
Router4 ↔ Multilayer Switch0	172.16.3.156/30
Router4 ↔ Multilayer Switch2	172.16.3.152/30

7.3 WAN / ISP Links

Each core router is also dual-homed to both ISPs (cross-connected in red on the topology diagram), providing full WAN redundancy — if either ISP or either router fails, connectivity is preserved:

Link	Network
ISP 1 ↔ Router3	195.136.17.0/30
ISP 1 ↔ Router4 (cross-link)	195.136.17.4/30
ISP 2 ↔ Router3 (cross-link)	195.136.17.8/30
ISP 2 ↔ Router4	195.136.17.12/30

8. VLAN Design

Each department is placed in its own VLAN, isolated from the others at Layer 2 and routed between at Layer 3 by the distribution-layer multilayer switches.

Benefits of this design include:

- Improved security through broadcast-domain and traffic isolation
- Reduced broadcast traffic and improved performance
- Easier day-to-day administration and troubleshooting
- Clear separation of departmental resources and policies

9. Core Technologies

The project implements the following technologies:

- OSPF (Open Shortest Path First) dynamic routing
- HSRP (Hot Standby Router Protocol) gateway redundancy
- VLAN segmentation
- Inter-VLAN routing
- Layer 3 switching
- PAT (NAT overload)
- DHCP with DHCP relay
- SSH version 2

10. OSPF Configuration

OSPF provides dynamic routing between all Layer 3 devices — the two core routers and the two distribution multilayer switches.

Configuration summary:

- Process ID: 1
- Area: 1 (single-area design, covering all core-to-distribution transit links)
- Loopback interfaces configured on each router to provide stable, predictable Router IDs

Benefits:

- Fast convergence after a link or device failure
- Automatic learning and propagation of routes
- Scalability for future growth of the network

11. HSRP Configuration

HSRP provides default-gateway redundancy for every VLAN, so that end devices always have a reachable gateway even if one distribution switch fails.

Switch	Role
Multilayer Switch0	Active router (higher priority)
Multilayer Switch2	Standby router

Features:

- Shared virtual gateway IP address per VLAN
- Automatic failover if the active switch goes down
- Preemption enabled, so the primary switch reclaims the active role once restored

12. NAT (PAT Overload)

Port Address Translation (PAT) translates many private internal IP addresses to a single public IP address per ISP uplink, allowing all internal hosts to reach the Internet simultaneously.

Advantages:

- Conserves public IPv4 address space
- Enables Internet connectivity for all internal VLANs
- Adds a layer of security by hiding internal addressing from external networks

13. DHCP Configuration

A centralized DHCP server, hosted in the Server Room VLAN at 172.16.3.245, issues addresses to every other VLAN using DHCP relay (IP helper-address) configured on each distribution-switch SVI, for example:

```
ip helper-address 172.16.3.245
```

Benefits:

- Centralized IP address management from a single server
- Automatic address assignment for all end devices
- Reduced administrative overhead compared to static addressing

14. SSH Security

SSH version 2 is configured on all network devices for secure remote management, replacing insecure Telnet access.

Security features include:

- Local username/password authentication
- Encrypted passwords (service password-encryption)
- VTY lines restricted to SSH transport only
- Telnet explicitly disabled on all VTY lines

15. Network Testing

The following tests were carried out in Packet Tracer simulation and real-time mode to validate the design:

Test	Status
PC-to-PC communication (intra-VLAN)	Passed
Inter-VLAN routing	Passed
DHCP address assignment	Passed
OSPF neighbor formation	Passed
HSRP failover	Passed
Internet access	Passed
PAT translation	Passed
SSH login	Passed

16. Results

The implemented solution successfully achieved all project objectives. The network demonstrated:

- High availability across the WAN edge and distribution layer
- Fast OSPF routing convergence after simulated link failures
- Secure remote administration via SSH
- Reliable, redundant Internet connectivity through dual ISPs
- Efficient VLAN segmentation across all six departments

17. Troubleshooting

Problems encountered during implementation, and how they were resolved:

- Incorrect VLAN assignments on access ports — corrected using show vlan brief and reassigning switchports
- OSPF neighbor adjacency failures — traced to mismatched subnet masks and hello/dead timers
- NAT configuration errors — resolved by correcting the ACL referenced by the NAT overload statement
- DHCP relay misconfiguration — fixed by verifying ip helper-address on every VLAN interface
- HSRP priority conflicts — resolved by explicitly setting priority and enabling preemption on the intended active switch

Each issue was resolved through systematic verification (show commands, ping/traceroute testing) and retesting.

18. Future Improvements

Potential enhancements for a future phase of this project include:

- EtherChannel for link aggregation between switches
- Rapid PVST+ for faster spanning-tree convergence
- Access Control Lists (ACLs) for granular traffic filtering
- IPv6 deployment alongside the existing IPv4 design
- A dedicated Wireless LAN Controller
- Centralized AAA (authentication, authorization, accounting) server
- SNMP monitoring and a Syslog server for centralized logging
- NTP for accurate, synchronized device clocks
- Quality of Service (QoS) policies for voice/video traffic

19. Conclusion

This project demonstrates the successful design and implementation of a modern enterprise network using Cisco technologies. The integration of OSPF, HSRP, VLANs, PAT, DHCP, SSH, and dual-ISP redundancy provides a secure, scalable, and fault-tolerant infrastructure suitable for real-world enterprise environments.

20. References

- Cisco Systems. Cisco Networking Academy CCNA Course Materials.
- Cisco IOS Configuration Guides.
- RFC 2328 – OSPF Version 2.
- RFC 2281 – Hot Standby Router Protocol (HSRP).
- RFC 3022 – Traditional NAT.
- RFC 1918 – Address Allocation for Private Internets.
- Cisco Packet Tracer Documentation.